

Proof Without a Verdict

Holder-Controlled Credentials, Limits on What May Be Asked,
and Tiered Issuers — An ACTIK Case

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October 2026

Preprint — comments welcome.

Keywords: verifiable credentials; selective disclosure; holder binding; credential revocation; recruitment; non-discrimination; digital certificates; employment records; trust lists; emerging economies; CamboVerse; Cambodia

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Abstract

Systems that verify certificates usually answer with a verdict: a tick, a green screen, the word *verified*. We argue that the verdict is the wrong output, and that the questions a verifier is asked to settle are set by the format before anyone looks at a credential. We make the argument with ACTIK, an open-source application for issuing and managing digital assets — certificates first, employment records second — built for Cambodia and intended as the data layer of the CamboVerse ecosystem.

Four design choices carry the argument. First, verification returns evidence rather than a verdict: who issued the credential, its standing as of a dated list, and the fields a reader must compare with the document in front of them; the result types contain no boolean. Second, an employer’s request for proof is written in a format that cannot ask for date of birth, sex, marital status, a photograph, a national identity number or a student number, and the limit is enforced three times — in the candidate’s application, in the database, and in the employer’s application — so that it does not depend on a registrar. Third, the signed trust list carries each issuer’s kind: a registered employer may sign employment records and nothing else, and every verifier says which kind of issuer it is. Fourth, withdrawal lists name no one, and a credential issued to a person who already holds a wallet is bound to that wallet’s key, so a copy or a replayed answer is refused.

The cryptography is standard and we claim none of it. Selective disclosure, key binding, status lists and limits on over-asking all exist. Cambodia’s national platform already verifies

official documents online, against a record it holds; what it does not do is put a credential in the holder’s own hand, from any accredited issuer, to be checked offline and shown in part. ACTIK does, and is offered beside the national platform as a choice, not in place of it. Our contribution is narrower than the mechanisms: where the limits sit, what the verifier is told, and which of the claims a design makes can be checked in its code. We separate claims that are true of the published system, claims carried from earlier work, claims that follow from stated premises, and guesses to be tested. Nothing has been issued to anyone.

1 Introduction

A student finishes a course on a digital topic and asks a simple question: can I have my certificate digitally — not a scan, not a link to someone else’s website, but a certificate I keep, that anyone can check, and that I can show without showing everything else? In Cambodia today the answer is mostly no. The national verification platform checks official documents against a record held on its own servers; it gives the holder a code that points there, not a credential of their own [2, 15]. A course from a training provider usually ends with a PDF or a sheet of paper.

Later the same person applies for a job. She brings a photocopy of her degree, certified as a true copy, and the employer files it. Nobody checks it against the university, because there is no easy way to; and the employer, if it wants to, can ask her age, whether she is married, and for a photograph, because the application form has spaces for them.

Two things are wrong in that scene, and they are usually treated separately. The first is that the document cannot be checked. The second is that the employer can ask for more than it needs. Technology for the first problem is mature: a signed credential can be checked by anyone, offline, in milliseconds. The second problem is usually left to law and to the employer’s conscience.

This paper is about a system that gives people their certificates digitally, and about the place where those two problems meet. When a candidate answers an employer with signed credentials instead of a form, the format of the request decides what can be asked, and the format of the answer decides what the employer is told. Both are design decisions, made once, and they shape every exchange that follows.

1.1 What already exists

We start by conceding most of the ground. Verifiable credentials are a W3C Recommendation [18]. Selective disclosure, in which the holder reveals some signed fields and withholds others, is an IETF standard, SD-JWT, which also specifies how a presentation is bound to the holder’s key [9, 13]. Status lists for revoking credentials without a per-credential lookup are a W3C Recommendation [19]. The European Union’s digital-identity framework already obliges the parties that request data from a wallet to register what they intend to ask for, and forbids them to ask for more [7, 8]. And Cambodia already has a national document-verification service, Verify.gov.kh, which places QR codes on diplomas and other official documents and checks them online against a record held by the platform [10, 11, 15].

None of that is ours. What the national platform does not offer — a credential the holder keeps and presents, from issuers beyond government bodies, checked without contacting anyone — is what ACTIK adds, and it is offered beside the platform, for people and institutions to choose, not instead of it (Section 8.2).

1.2 What we claim

We make four claims, and for each we say what is already known.

First: the output should be evidence, not a verdict. A verifier that says *verified* tells the reader that a registered key signed some bytes, and invites them to believe much more: that the person in front of them is the person named, that a two-week course and a four-year degree are equally weighty, that a credential checked against last month’s withdrawal list is current today. Research on security indicators has long found that people do not notice or act on the indicators designers rely on [17]. What we add is a concrete interface contract, enforced by the result types and by tests: the issuer first, its kind, the standing as of a dated list, the fields to compare, and three distinct ways of not succeeding.

Second: limits on what may be asked belong in the format. The EU mechanism limits over-asking through registration: a national register records what each requester intends to ask, and a certificate enforces it [7]. That requires a registrar, a register and a legal framework. We put a fixed limit in the request format itself, so that a request for date of birth, sex, marital status, a photograph, a national identity number or a student number cannot be written at all, and we enforce it at three points that do not trust each other. The limit is wider than the grounds listed in Cambodia’s Labour Law [14], and we say where and why (Section 2.4).

Third: the trust list should say what kind of issuer each issuer is. A trust list that admits an issuer usually admits it for everything. We sign each issuer’s kind into the list — an accredited institution may issue any credential type, a registered employer only employment records — and every verifier shows the kind beside the name. Employment records then carry their own rules: they cannot hold a salary, a reason for leaving or a performance rating, and *currently employed* is always shown as of the day it was signed.

Fourth: withdrawal should name no one, and a credential should be bound to its holder where it can be. Both are carried techniques. The withdrawal lists use the hashed-entry design of an earlier printed-certificate specification [2]; binding follows SD-JWT key binding [9]. What we add is their combination with the first three choices, and an account of what binding does not do.

Section 2 reviews what earlier work covers. Section 3 explains the method. Section 4 describes the system and what has been tested. Sections 5–7 develop the four claims. Section 8 discusses them, Section 9 lists what we do not know, Section 10 proposes what should happen next, and Section 11 concludes.

2 What Is Already Known

2.1 Credentials, selective disclosure and binding

The W3C data model defines a credential as a set of claims made by an issuer about a subject, and a presentation as data derived from credentials and shared by a holder [18]. SD-JWT signs, instead of each claim, a salted hash of it, so that a holder can later reveal a chosen subset; the verifier can check each revealed claim against the signed hashes without learning the rest [9]. The same standard defines key binding: the issuer places the holder’s public key in the credential, using the confirmation claim of RFC 7800 [13], and each presentation carries a short token signed with the matching private key, naming its audience and hashing the presentation it accompanies. ACTIK uses SD-JWT and its key binding as specified. We claim nothing about either.

2.2 Signed documents on paper

Signed payloads in QR codes are established: the EU Digital COVID Certificate put a COSE-signed, compressed, base45-encoded payload on paper at continental scale [6, 16]. The printed-certificate format ACTIK uses (Section 4.9) is KH-SQR Profile B, specified and implemented in an earlier preprint by one of the present authors [2, 3]. That work also supplies the withdrawal-

list entry design we carry: an entry is a hash of the issuer and the document number, so a list can be public without listing anyone.

2.3 Verification in Cambodia

Verify.gov.kh, the national platform, gives an issued document a QR code and a digital code which, with condensed information from the document, are stored on a blockchain; opening the code shows the stored contents, and the platform checks that the document comes from a recognised government body [15]. Press reports describe its use for grade-12 certificates and university degrees, including degrees of the university at which the present authors work, and its adoption beyond Cambodia [10, 11]. We have not verified those figures and do not rely on them.

Its architecture has been analysed in an earlier preprint by one of the present authors, and we carry that analysis rather than repeat it [2]. In short: the code on an issued document carries a *reference* to a record, a URL with a key in it, rather than the credential itself. Verification therefore depends on the platform staying reachable for the life of the document; a photograph of the document carries a working key; and official codes that open websites teach the public a habit that QR phishing exploits. The same analysis credits what a reference buys — a correction or withdrawal is visible to the next reader at once — and we return to that in Section 8.2.

2.4 Limits on what a requester may ask

Two bodies of rule bear on what an employer may ask a candidate. The first is data protection. Under the EU framework, a party that requests attributes from a digital-identity wallet must register its intended use in a national register, must not request more than it registered, and is identified to the wallet by a registration certificate [7, 8]. This is a limit by registration: flexible, per requester, and dependent on a registrar.

The second is labour law. Cambodia’s Labour Law, in its Article 12, prohibits an employer from taking account of race, colour, sex, creed, religion, political opinion, birth, social origin, or trade-union membership and activity in hiring and later employment decisions [14]. We could not reach the promulgated text from the environment in which this paper was prepared, and quote the list from secondary legal summaries; it should be checked before reliance. On that list, age and marital status do not appear. ACTIK’s request format excludes them anyway, together with a photograph, a national identity number, a student number, place of birth and contact details. That is a design choice, not a legal requirement, and Section 6 argues for it.

3 Method

3.1 What kind of paper this is

This is a design case with a conceptual argument. In Jaakkola’s terms it is closest to a theory adaptation: it takes ideas established for other purposes — credential formats, security-indicator research, data-protection limits on requesters — and applies them to a setting they were not built for [12]. It follows the method of a companion CamboVerse case by several of the present authors [5], including its four kinds of claim (Section 3.4).

3.2 How we use the case

Our case is ACTIK, an application whose purpose is to issue digital assets and to let their owners manage them. A digital certificate is the first kind of asset it handles; employment records are the second. ACTIK is planned as the data layer of CamboVerse, an open-source

Virtual Cambodia: under the agreed division, ACTIK holds identities, issued files, verification state and withdrawal, and CamboVerse displays what a person chooses to show [5].

We use ACTIK as an illustration, not as a test. Every design claim in Section 4 corresponds to code a reader can inspect and, in most cases, to a test that fails if the claim stops being true [1]. Nothing has been issued to a real person, no employer has used it, and no statement here is about how people behave with it.

3.3 What we checked before claiming anything

Before settling the claims we searched the standards and regulation named in Section 2. Three claims we had intended to make were withdrawn as a result. Key binding was not ours to claim: it is in SD-JWT [9]. Privacy-preserving revocation was not ours: status lists address it [19], and our entry design is carried from the printed-certificate work [2]. And limiting what a requester may ask was not ours either: the EU framework does it by registration [7]. What survives is the narrower claim stated in Section 1.2.

3.4 Four kinds of claim

Every substantive statement in this paper is one of four kinds [5]. A SPECIFIED claim is true of the published code and can be checked against it; where a test enforces it, we say so. A CARRIED claim is established in earlier work and imported. An ANALYTIC claim follows from stated premises; it holds if they do. A CONJECTURAL claim is a reasoned guess about how people will behave that nothing here establishes.

The line most easily blurred to our advantage is between analytic and conjectural. That a request format which has no field for marital status cannot carry a request for it is analytic. That employers who cannot ask for marital status will therefore discriminate less is conjectural — they can still ask at interview — and we have asked no employer.

3.5 Who we are

We are not independent of the case. One author directs the CamboVerse Center, an institute at the National University of Management; the other two are engineers on the CamboVerse project, one of whom, according to the repository history, wrote most of ACTIK’s original application. Parts of the system described in Section 4 were written with the assistance of an AI model, as the declarations record. The university at which we work issues degrees, and its degrees are verified through the national platform discussed in Section 2 [10]. These are affiliations and facts, not claims about who governs anything.

This cuts both ways. It strengthens Section 4: every statement there can be checked against public code and reproduced with public tests. It weakens Sections 5–8: we have an interest in the design looking important. The conjectural claims are where that interest could distort our judgement most, and they are exactly the ones we have not tested.

4 The System, Described

This section describes. It does not argue. Every statement in it is SPECIFIED.

4.1 Roles

Root. Holds an offline signing key whose public half is built into the application. It signs the trust list. Approving an issuer in the database does not make it trusted; the Root’s next signed list does.

Accredited institution. An issuer on the trust list that may issue any credential type.

Registered employer. An issuer on the trust list that may issue employment records only.

Holder. A person with a wallet. The wallet is end-to-end encrypted: the database holds ciphertext, and only the holder’s PIN or passkey opens it.

Requester. Any signed-in account that asks holders for proof — in practice, an employer recruiting.

4.2 Issuing and accepting

An issuer signs an SD-JWT credential in its own browser with a key held in memory as a non-extractable Web Crypto key [20]; the key never leaves that object. The credential goes to an outbox addressed to the recipient’s email. The database admits it only if the signed-in user owns an issuer of the right kind for that credential type. Nothing reaches the recipient’s wallet until they accept it, and they may decline, which deletes the offer. On acceptance the holder’s application verifies the credential against the trust list and the issuer’s withdrawal list, and stores it encrypted.

4.3 The trust list and issuer kinds

The trust list is a statement signed by the Root, valid for at most 31 days, listing each issuer’s decentralised identifier, name, kind and keys, with each key’s validity window and status. A verifier refuses a list signed by any other key, an expired list, or a list older than one it has already seen. A key that has been retired continues to vouch for what it signed before retirement; a revoked key vouches for nothing. A registered employer that signs any credential other than an employment record is refused with the reason `TYPE_NOT_ALLOWED_FOR_ISSUER`, in the application and on printed codes. The build step that the Root runs prints each issuer’s kind for review and refuses to sign a kind it does not recognise.

4.4 Withdrawal

Each issuer signs its own withdrawal list with a key the trust list accepts for it. An entry is the SHA-256 hash of a domain tag, the issuer and either the document number or the credential’s identifier, with one of two fixed reasons — *withdrawn* or *replaced by a corrected credential* — and a date. The database refuses a list with any other field, so a document number or a free-text explanation cannot be published by mistake. A verifier reports a credential’s standing as *clear* against a named list version and date, *unchecked* if the issuer publishes no list or its list has lapsed, or *withdrawn*, with the reason.

4.5 Holder binding

The first time a wallet is unlocked, the application makes a holder key pair. The private half is encrypted with the wallet’s own key before it is stored, and is held in memory as a non-extractable key. When a recipient already has a wallet, the issuer writes the holder’s public key into the credential as a confirmation claim [13]. Every presentation of a bound credential must then carry a key-binding token signed by that key, naming its audience — one share link or one proof request — and hashing the presentation [9]. A verifier refuses a bound credential presented without the token, with a token from any other key, with a token dated in the future or not matching the disclosures, or with a token made for another audience. A credential issued before the recipient had a wallet is unbound, still verifies, and is labelled as unbound.

Table 1: What a proof request can ask for. Every answer discloses the fields in the middle column; a request may add those in the right column. Nothing else can be requested. (SPECIFIED; enforced by `test-proof.ts` and by the database guard.)

Credential type	Always disclosed	May also be asked
Academic degree	name, institution, degree, graduation date, certificate number	major, GPA
Professional certification	name, institution, certification, issuing body, date, licence number, expiry	—
Certificate of completion	name, institution, type, programme, completion date	duration, department or role
Attendance or participation	name, institution, type, event, date, organiser	role
Merit or excellence award	name, institution, achievement, date awarded	basis of award
Appreciation or service	name, institution, type, reason, date	capacity
Employment record	name, employer, job title, employment type, start, status, end	department, role

Never requestable: date of birth or age, sex or gender, marital status, photograph, national identity number, student number, place of birth, religion or ethnicity, email or phone number.

4.6 Sharing and verifying

A holder can share a credential through a link. The share discloses a fixed set of fields — the holder’s name, the issuer, the document number and date, and the substance of the credential — plus any optional fields the holder selects, and can be limited to a single view, enforced by the database. The verifier’s page checks the presentation on the verifier’s device and shows the issuer first, its kind, the standing, the four fields to compare with any paper document, and whether the presentation was bound.

4.7 Proof requests

A requester writes a request naming up to five credential types, each with optional extra fields from a short list for that type, and an expiry of at most 90 days, and shares its link. Table 1 gives the list. A request cannot name any field outside it. A candidate opens the link, signs in, and sees the matching credentials in their wallet, each already checked; a withdrawn one cannot be chosen. The candidate sees exactly what the requester will see before sending, and chooses the contact details they leave. The answer can be withdrawn by the candidate, never edited, and is visible only to the candidate and the requester. Nobody can list requests or browse candidates.

The limit is enforced three times. The candidate’s application discloses only allowed fields, whatever the request says. The database decodes every disclosure in every answer and refuses an answer that carries any other claim, or a credential of the wrong type. The requester’s application refuses to display an answer that discloses anything else — not even its allowed part.

4.8 Employment records

An employment record carries the holder’s name, the employer, a job title, an employment type, a start date and a status — *current* or *ended*, with an end date — and optionally a department and a one-line role. The issuing form has no field for salary, reason for leaving, performance or discipline, and no free-text note or attached file; the database decodes each record and refuses any claim outside that set. *Current* is displayed everywhere as *current, as of* the date the

Table 2: Automated test suites (`npm test`). Counts are of assertions on the cited branch [1].

Suite	Assertions	What it holds to
<code>test-sdjwt.ts</code>	21	issuance, selective disclosure, signature and expiry checks
<code>test-claim.ts</code>	11	nothing enters a wallet unless the trust list’s key signed it
<code>test-trust.ts</code>	48	Root-signed trust list, rollback, key windows; withdrawal lists that name no one
<code>test-printed.ts</code>	48	the thirteen Profile B vectors; printed codes on the trust layer
<code>test-museum.ts</code>	19	exhibit export carries no original file and no verdict
<code>test-proof.ts</code>	34	what can be asked, what is disclosed, how an answer is checked
<code>test-employment.ts</code>	21	issuer kinds; what an employment record may say
<code>test-binding.ts</code>	15	key binding: missing, foreign, replayed and tampered proofs
Total	217	

employer signed it. When a job ends, the employer issues a new record and withdraws the old one as replaced.

4.9 Printed certificates

For degrees and professional certifications that carry a document number, the issuer also signs a printed code in the KH-SQR Profile B format: CBOR claims in a COSE signature, compressed and base45-encoded behind a `KH1:` prefix [2, 16]. The code is delivered to the holder with the credential, checked on acceptance, and kept in the wallet for reprinting. ACTIK’s scanner verifies it offline against the same trust and withdrawal lists. The vendored implementation passes all thirteen Profile B verification vectors of the original specification.

4.10 The seam with CamboVerse

A holder can export a credential as an exhibit for a CamboVerse personal museum. The exhibit carries the SHA-256 of the original file, a picture the holder has cropped and covered, reduced and re-encoded on their device, the printed code if the holder chooses, and the withdrawal-list hashes under which the credential would appear. It carries no account identity, no original file and no *verified* flag; its visibility defaults to private, and CamboVerse is to show the stricter of the exhibit’s and the room’s setting.

4.11 What is tested and measured

Table 2 lists the automated suites. Database rules were exercised against a local PostgreSQL 16 instance with the platform’s authentication schema stubbed, and interface behaviour in headless Chromium against the built application with a stand-in for the hosted database; neither is the hosted service. Table 3 reports sizes and timings produced by a script in the repository.

4.12 What the system does not show

There is no deployment. No credential has been issued to a real person, no employer has written a request, and no candidate has answered one. The browser checks ran against a stand-in database, not the hosted service. No timing has been taken on the low-cost phones the design targets. The Root role has been exercised only in a rehearsal of its key ceremony; nobody holds it.

Table 3: Sizes and timings (`npm run measure`), with fresh keys and fictional names. Timings are of Node.js 22 on a four-core 2.8 GHz server processor, not of a phone, and vary by about 0.2 ms between runs; they are illustration, not a claim about the devices the design targets.

Quantity	Value
Degree credential, all claims (bytes)	1,909
Default share presentation (bytes)	1,702
Key-binding token added to it (bytes)	366
Trust list with 100 issuers (bytes)	34,886
Withdrawal list with 1,000 entries (bytes)	118,135
Printed code, Khmer holder name (characters)	417
Verify a bound share against a 1,000-entry list, median / 95th percentile (ms)	1.7 / 2.4
Open a 100-issuer trust list, median / 95th percentile (ms)	1.2 / 1.7
Verify a printed code, median / 95th percentile (ms)	9.0 / 12.3
Application shell cached for offline use (KiB)	1,392

5 Evidence, Not a Verdict

5.1 What a verdict hides

A signature check establishes one fact: a key the verifier trusts signed these bytes. Everything else a reader wants to know lies outside it. The paper document may be a forgery carrying a genuine code copied from another certificate. The person presenting may not be the person named. The issuer may have withdrawn the credential after the verifier’s last list. The issuer may be a three-person company whose records weigh little, or a university. A single *verified* mark collapses all of these into one bit.

Proposition 1 (ANALYTIC). *If a verification interface reports a single success state, then a credential copied onto a forged document, a credential presented by someone other than its holder, a credential withdrawn after the last list, and a credential from a low-weight issuer are reported identically to a credential with none of those defects.* It follows from the premise that the signature check is the same in all five cases.

5.2 What ACTIK reports instead

ACTIK’s result types make the verdict hard to write. A successful check returns a record, not a flag: the issuer and its kind from the signed list, the standing with its list version and date, the four fields a reader must compare with any document (name, document number, issuing organisation, date), and whether the presentation was bound to its holder (SPECIFIED). A failure is one of three distinct outcomes: *refused*, with a stable reason; *could not check*, when the verifier’s own trust state is at fault and the credential is not; and *withdrawn*, naming the issuer’s decision (SPECIFIED). Security indicator research suggests that a bare success mark is not read the way designers hope [17] (CARRIED).

Proposition 2 (CONJECTURAL). *Readers shown the issuer, its kind and the dated standing first will distinguish a short course from a degree, and an unchecked standing from a clear one, more often than readers shown a single success mark.* Nothing here tests it.

6 Asking Less, by Design

6.1 Limits by registration and limits by format

A limit on over-asking can live in two places. It can live in a register: each requester declares what it will ask, and the wallet refuses more. That is the EU mechanism [7] (CARRIED). Or it

can live in the format: the request language has no way to express the forbidden field. ACTIK does the second (SPECIFIED).

Proposition 3 (ANALYTIC). *A limit written into the request format binds every requester without a registrar, at the cost of being the same for every requester. A register can allow a bank to ask what an employer cannot. A format cannot. We accept that cost because the only requester ACTIK is built for is an employer judging qualifications, and because Cambodia has no register of requesters to rely on.*

6.2 Why three enforcement points

A limit enforced only in the candidate’s application is defeated by a modified application; one enforced only in the database is defeated by a misconfigured database; one enforced only in the requester’s application is defeated by reading the database directly. ACTIK enforces the allowlist at all three (SPECIFIED). This is an instance of a more general observation made in an earlier Cambodia case by several of the present authors: a control enforced in one layer does not travel to claims made in another [4] (CARRIED, self-cited).

6.3 Why wider than the law

The excluded fields are wider than the grounds in Article 12 of the Labour Law [14]: age, marital status and a photograph are not on that list as we have it. We exclude them because each is routinely asked for in job advertisements, none bears on whether a credential is genuine, and a photograph discloses sex, apparent age and ethnicity at once. The exclusion of a student number and a national identity number has a different reason: they are identifiers that join records across systems, and a qualification can be judged without them.

Proposition 4 (CONJECTURAL). *Employers who receive candidates’ qualifications as signed answers without age, sex, marital status or a photograph will shortlist on qualifications more than employers who receive paper applications.* The opposing conjecture is as reasonable: that employers will ask the excluded questions at interview, and that the format changes the order in which bias acts, not whether it acts. The name, which the format must disclose so that identity can be checked, itself carries signals of sex and ethnicity. We have not tested either conjecture.

7 Who May Say What

7.1 Issuer kinds in the signed list

A trust list that admits an issuer usually admits its key for any statement. If employers are to issue employment records, that is a problem: a key admitted so that a small company can confirm its staff should not let it confirm a degree. ACTIK signs each issuer’s kind into the list and refuses any other credential type from an employer (SPECIFIED).

Proposition 5 (ANALYTIC). *If issuer kinds are recorded only in a database, then whoever can write that database can promote an issuer; if they are signed into the list, only the Root can.* It follows from the premise that verifiers trust the Root’s signature and not the database.

7.2 Records people will carry

An employment record is a statement about a person made by someone with power over them. ACTIK’s record cannot carry salary, reason for leaving, performance or discipline, and the employee must accept it before it reaches their wallet (SPECIFIED).

Proposition 6 (CONJECTURAL). *Employees will accept and present employment records more readily when the format guarantees that a record cannot carry anything that could be used against them.* The cost is that a record says less than a reference letter does. We think that is the right side to err on, and we have not asked anyone.

7.3 “Current” has a date

A record saying *currently employed* is true on the day it is signed and becomes false without anyone touching it. ACTIK never displays it without that date (SPECIFIED), and asks employers to replace and withdraw a record when the job ends. A verifier is told how fresh the withdrawal list it checked is, and offline verifiers can be up to the list’s refresh interval behind the issuer (ANALYTIC).

7.4 Withdrawal and binding

The withdrawal list names no one: each entry is a hash, so a reader learns nothing unless they already hold the document number or credential identifier it was computed from (CARRIED [2]). Where document numbers are sequential, a reader can hash candidates and learn *that* a number was withdrawn, never whose it was or why (ANALYTIC). Binding removes two attacks: presenting a copied credential, and replaying an answer made for one requester to another (SPECIFIED, tested). It does not remove collusion: a holder who shares their PIN, or hands over an unlocked phone, can have someone else present for them (ANALYTIC). That is why the name is still shown and an identity check at interview is still advised.

8 Discussion

8.1 What the paper adds

The parts are known; the paper adds an arrangement and an account of it. A verifier contract without a verdict, enforced by types and tests. A fixed limit on what may be asked, written into the request format and enforced at three points, suited to a setting without a register of requesters. Issuer kinds signed into the trust list, and an employment record designed so that it cannot be used against its holder. And a statement, for each, of which claims are true of the code and which are guesses.

8.2 Beside the national platform: more, and a choice

ACTIK is not proposed against Verify.gov.kh. It is proposed as a second option that does more of what a holder needs, for people and institutions to choose between. Table 4 sets the two side by side. The rows for Verify.gov.kh rest on its own published description [15] and on the earlier analysis [2] (CARRIED); the rows for ACTIK are SPECIFIED.

Two rows favour the national platform and we state them plainly. A withdrawal on an online record takes effect at the next lookup, where ACTIK’s takes effect only when a verifier next refreshes the issuer’s list. And Verify.gov.kh is an operating government service with institutional standing; ACTIK is a tested design that nobody yet uses.

The choice is not exclusive. A university may keep issuing through the national platform and also sign an ACTIK credential for the same degree; a training provider whose courses the national platform does not cover can issue through ACTIK alone; a holder may present whichever the verifier in front of them accepts. We propose no integration between the two and claim none.

Proposition 7 (CONJECTURAL). *Given the choice, holders of short courses and employment records, for which no national record exists, will take up a holder-kept credential more readily than holders of degrees, which the national platform already covers.* Nothing here tests it.

8.3 The ecosystem

Within CamboVerse, ACTIK is the layer that holds data and CamboVerse the layer that shows it, joined by a portable file the holder exports rather than a live interface. This keeps documents

Table 4: What each option gives a holder and a verifier. Neither is required by the other; an issuer may use both, and a holder may present whichever a verifier accepts.

	Verify.gov.kh	ACTIK
What the holder has	a document with a code that points to the platform’s record	a signed credential in their own encrypted wallet, and optionally a printed code that carries the credential itself
Who can issue	recognised government bodies, as the platform describes itself	any issuer the Root admits: accredited institutions, including training providers, and registered employers for employment records
How it is checked	online, against the platform’s servers	offline, against signed lists the verifier already holds
What is shown	the stored contents of the record	what the holder chooses to disclose, above a fixed minimum
Proof that the presenter is the holder	none in the code; a copy of the link works	bound credentials need the holder’s key, for that verifier only
Withdrawal	visible at once to the next reader	visible as of the issuer’s last signed list, up to its refresh interval later
Standing	a national government service in operation	a design with tests, not deployed

and personal data out of the display layer and means that the display layer never needs to know who looks at whose exhibit. The same arrangement — a holder-controlled data layer, a display layer that verifies what it is given, and a file between them — is available to any platform that wants to show people’s achievements without holding them (ANALYTIC).

9 What We Do Not Know

We would rather list the paper’s limits than have them found.

There is no empirical content about people. Propositions 2, 4, 6 and 7 are conjectures. Propositions 1, 3 and 5 follow from stated premises and are not thereby true of the world.

There is no deployment. The database rules were tested against a local instance with a stubbed authentication schema, and the interface against a stand-in for the hosted database. Neither is evidence about the hosted service.

No measurement was taken on a low-cost phone. Table 3 reports a server processor. The design targets phones on mobile networks, and a 1,000-entry withdrawal list is about 118 KB, which grows linearly with withdrawals; we have not measured what that costs on the devices that matter.

The Root is a role nobody holds. Who should hold it, how institutions and employers are admitted, and who republishes the list every month are governance questions this paper does not answer. With many issuers, a verifier is in the end trusting whoever admits them.

The labour-law list in Section 2.4 is quoted from secondary summaries. Holder binding covers only credentials issued after the holder had a wallet; earlier ones stay unbound. And the Khmer text of the interface has not yet been reviewed by native speakers.

10 What Should Happen Next

Four studies would move this materially, in the order we would run them.

First, a measurement on two or three low-cost Android phones on a mobile network: time to open the trust list, verify a bound share, verify a printed code, and fetch a withdrawal list of realistic size. It can be run now, needs no participants, and would replace Table 3’s server figures with the ones that matter.

Second, a reading study of the verification screen: participants shown either ACTIK’s evidence layout or a single success mark, across a short course, a degree, a withdrawn credential and an unchecked one, asked what they would conclude. It tests Proposition 2 directly.

Third, structured interviews with a small number of employers and registrars about the request format and employment records: what they would ask that the format forbids, and whether employees would accept records under these rules. It bears on Propositions 4 and 6 without claiming to test them.

Fourth, a pilot with one issuer and one employer who already hires its graduates, once the governance of the Root role is settled. Only a pilot can say whether any of this changes who gets hired.

11 Conclusion

Someone who finishes a course should be able to keep the certificate, show it to anyone, and show only what is needed. Cambodia’s national platform verifies official records online; ACTIK puts a signed credential in the holder’s hand, and is offered beside the platform for people to choose. A certificate check that says *verified* answers a narrower question than it appears to, and a request for proof that can ask anything will be used to ask too much. Both are settled by format before anyone looks at a credential. ACTIK’s answer is to return evidence instead of a verdict, to write the limits on asking into the request itself, to sign into the trust list what kind of issuer each issuer is, and to bind credentials to their holders where it can. None of the mechanisms is new. Their arrangement is, and so is the discipline of saying, for each claim, whether the code makes it true or whether it is a hope. The code is public; the hopes are untested.

Declarations

Conflict of interest. The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest. All authors hold appointments at the National University of Management. The first author directs the CamboVerse Center, an institute at that University; the second and third are engineers on the CamboVerse project. These are affiliations. The University issues degrees and uses the national verification platform discussed in Section 2. Participation here is as individual researchers; the paper is not University output, was not reviewed or approved under any University process, and should not be cited as a University position.

Funding. This research received no external funding.

Author contributions. *To be confirmed by the authors before posting.* SC conceived the study and the claims, and drafted the manuscript. MV built the ACTIK application on which Section 4 reports, as recorded in the repository history, and verified the description of it. LK [contribution to be stated by the authors]. All authors reviewed and approved the submitted version and accept responsibility for its content.

Relationship to prior work. The printed-certificate format and the withdrawal-list entry design are carried from a preprint by the first author [2]; the method, including the four kinds of claim, from a companion CamboVerse case by several of the present authors [5]; and the observation about controls in layers from a further companion paper [4]. None of them is a source of evidence for the claims made here.

Data and code availability. No data about people were generated or analysed. The source code, tests, database schema and measurement script are published at <https://github.com/camboversecenter/ACTIK> [1]. Every SPECIFIED claim in Section 4 can be checked against it, the counts in Table 2 are reproduced by `npm test`, and the figures in Table 3 by `npm run measure`. Test fixtures use fictional people; some fixtures and documentation name real institutions as illustrations only, which implies no relationship with them.

Generative AI disclosure. The authors used Anthropic’s Claude during this work: to implement parts of the system described in Section 4 (the trust layer, withdrawal lists, printed certificates, proof requests, employment records, holder binding and their tests, as recorded in the repository history), to search for prior art, to verify reference metadata, and to draft and typeset this manuscript. The search for prior art withdrew three claims we had intended to make (Section 3). The authors reviewed the AI-assisted code and text, checked the cited sources, and take full responsibility for the content of this paper.

Disclaimer. This is a preprint and has not been peer reviewed.

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